Lily Poursoltan

Ph.D. Candidate in Information Systems
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Education

University of California San Diego, Rady School of Management *2022 – Present*
Ph.D. Candidate in Information Systems (Innovation, Technology & Operations Area) *Expected Dec 2026*

Dissertation: Essays on the Design, Development, and Evaluation of Intelligent Work Systems: Human–AI Collaboration, Delegation, and Governance in Healthcare

Dissertation Committee: Kevin Zhu (Chair), Karandeep Singh (Co-Advisor), Terrence August, and Hyoduk Shin

Queen's University, Smith School of Business *2020 – 2021*
M.Sc. in Management Analytics (with distinction)

Focus: Applied Artificial Intelligence, Machine Learning, and Business Analytics

Research Interests

Topics: Human–AI Collaboration & Delegation; Intelligent Work Systems & Organizational Design; AI Adoption, Evaluation & Governance; AI Agents & Agentic Systems; Healthcare IT

Methodologies: Econometrics & Causal Inference; Machine Learning; NLP & LLM-Based Measurement

Job Market Paper

L. Poursoltan, W. Chen, K. Singh, K. Zhu, "Human-in-the-Loop Generative AI for Knowledge Work: Interaction-Level Task–Technology Fit and Selective Deployment," revise & resubmit at ***Information Systems Research***.

Best Student Paper Award, 2026 Conference on Health Information Technology and Analytics (CHITA)

Selected Peer-Reviewed Publications

L. Poursoltan, J. Cao, W. Chen, E. Kim, T. K. Nakai, A. Boussina, A. Chua, J. Pan, M. Millen, M. Tai-Seale, C. A. Longhurst, K. Zhu, K. Singh, "Physician Edits to AI-Drafted Patient Messages and Their Impact on Clinical Workload," ***NEJM AI***, Vol. 3, No. 9, 2026. [paper]

(NEJM Group; publisher of *The New England Journal of Medicine*; 2025 Journal Impact Factor: 84.5)

Selected Media Coverage: *Healthcare Innovation* | *2 Minute Medicine*

L. Poursoltan, J. Cao, B. Clay, B. Trimble, L. Adrid, J. Pan, A. Chua, J. Bell, C. A. Longhurst, K. Zhu, K. Singh, "Prospective Comparison of Econometric, Machine Learning, and Foundation Models for Forecasting Emergency Department Boarding Patients," ***npj Health Systems (Nature Portfolio)***, Vol. 2, Article No. 49, 2025. [paper]

L. Poursoltan, A. Boussina, J. Cao, E. Kim, P. S. Bedmutha, H. Majmudar, M. Ramsis, A. Khalessi, K. Singh, "Generating Real-World Evidence from AI Deployed in Health Systems," ***Learning Health Systems***, forthcoming.

Working & Under-Review Papers

A. Noori,., **L. Poursoltan**, et al., "A Global Log for Medical AI," minor revision (third round) at ***Nature***. [manuscript]

L. Poursoltan, K. Singh, M. Tai-Seale, K. Zhu, "From Adoption to Productivity: How Clinicians Use EHR-Integrated Generative AI," soon to be submitted to ***Information Systems Research***.

L. Poursoltan, E. Kim, J. Cao, A. Bhattacharya, A. Boussina, A. Chua, J. Pan, R. El-Kareh, C. VanDenBerg, C. Longhurst, K. Singh, "Automating Analysis of Health System Incident Reports with Large Language Models," under review at ***npj Digital Medicine (Nature Portfolio)***.

L. Poursoltan, K. Singh, K. Zhu, "Scaling Patient Outreach with AI Agents: Work Reconfiguration and Operational Outcomes," working paper, target: ***Management Science***.

Other Peer-Reviewed Publications

N. Ekbote, M. Akhondzadeh, R. Graham, J. Bell, D. Glasser, R. El-Kareh, A. Chua, A. Boussina, A. Tai-Seale, **L. Poursoltan**, K. Singh, P. Agnihotri, M. Tai-Seale, "Large Language Models for Summarizing Advance Care Planning Information from Goals of Care Notes in the EHR," ***Learning Health Systems***, Vol. 10, Supplement 1, Article No. e70086, 2026. [paper]

L. Poursoltan, S. M. Seyedhosseini, A. Jabbarzadeh, "Green Closed-Loop Supply Chain Network under the COVID-19 Pandemic," ***Sustainability***, Vol. 13, No. 16, Article No. 9407, 2021. [paper]

L. Poursoltan, S. M. Seyedhosseini, A. Jabbarzadeh, "A Two-Level Closed-Loop Supply Chain under the Contract of Vendor-Managed Inventory with Learning: A Novel Hybrid Algorithm," ***Journal of Industrial and Production Engineering***, Vol. 38, No. 4, pp. 254–270, 2021. [paper]

L. Poursoltan, S. M. Seyedhosseini, A. Jabbarzadeh, "An Extension to the Economic Production Quantity Problem with Deteriorating Products Considering Random Machine Breakdown and Stochastic Repair Time," ***International Journal of Engineering***, Vol. 33, No. 8, pp. 1567–1578, 2020. [paper]

A. Golmohammadi, H. Rasay, Z. Sobhani, **L. Poursoltan**, "Modeling and Analyzing One Vendor–Multiple Retailer VMI Supply Chain Using Stackelberg Game Theory," ***Industrial Engineering & Management Systems***, Vol. 15, No. 4, pp. 385–395, 2016. [paper]

Research Funding & Grants

- **A Disease-Agnostic Supervisory Agent for Clinical AI Safety ($4.5M)** *2026*
  Advanced Research Projects Agency for Health (ARPA-H), ADVOCATE Program
  Role: Research Contributor (including proposal development), UC San Diego Health; PI: Karandeep Singh
- **AIVIS: AI-Based Clinical Decision Support for Sepsis ($1.26M)** *2023–2025*
  National Institutes of Health (NIH/NIAID), R42AI177108
  Role: Research Contributor (including proposal development), UC San Diego Health; Site PI: Karandeep Singh
- **Artificial Intelligence and Predictive Analytics in Healthcare** *2022–2023*
  Joan & Irwin Jacobs Center for Health Innovation, UC San Diego Health
  Role: Graduate Research Fellow

Research Appointments

Data Scientist – Jacobs Center for Health Innovation, UC San Diego Health *Sep 2022 – Present*

Develop, deploy, and evaluate predictive, generative, and agentic AI systems using EHR and operational data, focusing on human–AI collaboration, technology adoption, automation, clinical workflows, and productivity.

Data Scientist – Scotiabank Centre for Analytics & AI, Smith School of Business *Jul 2021 – Sep 2022*

Conducted applied NLP and machine-learning research on large-scale customer call transcripts to identify behavioral patterns, customer interests, and business-relevant insights.

Awards & Honors

- **Best Student Paper Award** – Conference on Health Information Technology and Analytics (CHITA) *2026*
- **D.I. McLeod Dean's List Scholarship** *2020–2021*
- **Dean's Award for Academic Excellence** – Iran University of Science and Technology *2020*
- **National Elites Foundation Award** *2020*
- **Excellence in Teaching Award,** Graduate and Undergraduate Levels – Azad University *2018*

Invited Talks

Duke University, Durham, NC, 2026

University of California, San Francisco (UCSF), San Francisco, CA, 2026

University of Southern California, Los Angeles, CA, 2026

Conference Presentations

Conference on Information Systems and Technology (CIST), San Francisco, California, 2026 *(Scheduled)*
INFORMS Annual Meeting, San Francisco, California, 2026 *(Scheduled)*
Wharton Business and Generative AI Conference, San Francisco, California, 2026
Machine Learning for Healthcare (MLHC), Johns Hopkins University, Baltimore, Maryland, 2026
Conference on Health IT and Analytics (CHITA), Johns Hopkins Carey Business School, Washington, 2026
Production and Operations Management Society (POMS) Annual Conference, Reno, Nevada, 2026
Symposium on Artificial Intelligence in Learning Health Systems (SAIL), Río Grande, Puerto Rico, 2026
Conference on Information Systems and Technology (CIST), Atlanta, Georgia, 2025
INFORMS Annual Meeting & Workshop on Data Science, Atlanta, Georgia, 2025
Wharton Business and Generative AI Conference, San Francisco, California, 2025
Conference on Health IT and Analytics (CHITA), McCombs School of Business, UT Austin, Texas, 2025

Doctoral Consortia

INFORMS ISS Doctoral Consortium, San Francisco, CA, 2026 *(Selected Participant; Scheduled)*
AMCIS Doctoral Consortium, Reno, NV, 2026 *(Selected Participant; Senior PhD Panelist)*
CHITA Doctoral Consortium, JHU Carey, Washington, DC, 2026 *(Selected Participant)*

Teaching Experience

- **Instructor**
  - **Operating Successful Business,** Undergraduate, UC San Diego *Summers 2025 & 2026*
    Integrated applied analytics, Python, and AI-assisted coding; taught in both in-person and online formats. Student ratings: 4.82/5 (in person) and 4.78/5 (online).
- **Curriculum Development**
  - **Generative AI for Business,** UC San Diego
    Contributed curriculum and instructional materials on LLM foundations, prompt engineering, retrieval-augmented generation (RAG), AI agents and agentic systems, fine-tuning, model evaluation, and AI safety and ethics, with emphasis on business applications, implementation, and evaluation.
- **Selected Guest Lectures and Teaching Assistant Experience**
  - **AI & Disruptive Technology Strategy,** EMBA/MBA, UC San Diego *Winter & Fall 2025*
    Delivered guest lectures, led review sessions and office hours, and facilitated case discussions on applied AI, generative AI, and LLMs, emphasizing technology adoption, implementation, value creation, and governance.
  - **Business Analytics,** Master of Science in Business Analytics (MSBA), UC San Diego *Fall 2024*
    Led weekly review sessions and office hours on applied analytics, statistical modeling, and programming in Python and R.
  - **Supply Chain Analytics,** MSBA, UC San Diego *Spring 2024 & 2025*
    Led review sessions and office hours on forecasting, optimization, and applied analytics
  - **Operations, Information Systems & Data Analysis,** MBA, UC San Diego *Winter 2023*
    Led review sessions and office hours on applied data analysis, information systems, quantitative decision-making, and using information to improve organizational processes and performance.
  - **AI in Marketing,** MMAI, Queen's University, Smith School of Business *Winter 2021*

Supported review sessions and office hours on applied machine learning and AI methods for data-driven marketing and decision-making.

- **Instructor**
  - **Applied Probability and Statistics; Supply Chain Management & Analytics**

    Iran University of Science and Technology; Azad University, North Tehran Branch *2016 – 2018*

    Taught applied statistics, quantitative modeling, optimization, and quantitative decision-making; developed course materials for classes of approximately 60–150 students per semester. Student ratings: 4.89–4.95/5.

Professional Service

Ad-hoc Referee – ICIS; CIST; Information Systems Frontiers; npj Health Systems

Judge – 1st Annual Physician AI Hackathon, UC San Diego *2026*

PhD Peer Mentor – Rady School of Management, University of California San Diego *2024 – 2026*

Mentored first-year PhD students on program structure, departmental culture, and academic life.

Women in Data Science Ambassador – Stanford University *2020 – Present*

Technical Skills

AI & Machine Learning: Large Language Models (LLMs) & Evaluation, Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), Fine-Tuning & Low-Rank Adaptation (LoRA), Agentic AI, Predictive Modeling, Time-Series Forecasting, Optimization

Programming: Python, R, SQL, MATLAB, Java

Technical Training: Data Science Certificate, University of Toronto